EM POWER & POYNTING VECTOR

Time-avg power density = real part of complex Poynting vector: $\langle\vec{S}\rangle = \frac{1}{2}\mathrm{Re}(\vec{E}\times\vec{H}^*)$ [W/m^2], radially outward $\hat{r}$

Plane wave free space: $\langle S\rangle = |E|^2/(2\eta_0) = |E|^2/240\pi$

$\eta_0 = Z_0 = \sqrt{\mu_0/\varepsilon_0} \approx 120\pi \approx 377\,\Omega$; $k_0 = 2\pi/\lambda = \omega/c$

Key: $\vec{E} = E_\theta\hat{\theta}$, $\vec{H} = H_\phi\hat{\phi} \Rightarrow \hat{\theta}\times\hat{\phi} = \hat{r}$ (power flows outward)

Elementary dipole full fields (length dl, current I, along $\hat{z}$):

$$H_\phi = \frac{I dl \sin\theta}{4\pi}\left(\frac{jk}{r} + \frac{1}{r^2}\right)e^{-jkr}$$

$$E_\theta = \frac{I dl \sin\theta\, Z_0}{4\pi k}\left(\frac{jk^2}{r} + \frac{k}{r^2} - \frac{j}{r^3}\right)e^{-jkr}$$

Expanding $E_\theta H_\phi^*$: all intermediate $1/r^3, 1/r^4$ terms cancel $\Rightarrow$ $\langle S_r\rangle = \frac{1}{2}\mathrm{Re}(E_\theta H_\phi^*) = \frac{Z_0 k^2 I^2 dl^2 \sin^2\theta}{32\pi^2 r^2}$ (only real k^3/r^2 survives)

Far-field only ($r \gg \lambda$): $E_\theta = \frac{jkZ_0 I dl \sin\theta}{4\pi r}e^{-jkr}$, $H_\phi = E_\theta/Z_0$

TOTAL REAL POWER & RADIATION RESISTANCE

$P_{\mathrm{rad}} = \oint\langle\vec{S}\rangle\cdot d\vec{A} = \int_0^{2\pi}\int_0^{\pi} U(\theta,\phi)\sin\theta\, d\theta\, d\phi$

Use $\int_0^\pi \sin^3\theta\, d\theta = 4/3$:

Elementary dipole: $P_{\mathrm{rad}} = 40\pi^2 I^2 (dl/\lambda)^2$ W, $R_{\mathrm{rad}} = 80\pi^2(dl/\lambda)^2$

Short dipole (triangular current): $R_{\mathrm{rad}} = 20\pi^2(L/\lambda)^2$, $D = 1.5$, $h_{\mathrm{eff}} = L/2$

Half-wave dipole: $R_{\mathrm{rad}} \approx 73\,\Omega$, $D \approx 1.64$, HPBW= 78°, $A_{\mathrm{eff}} = 0.13\lambda^2$

General: $P_{\mathrm{rad}} = \frac{1}{2}|I_{\mathrm{feed}}|^2 R_{\mathrm{rad}} \Rightarrow R_{\mathrm{rad}} = 2P_{\mathrm{rad}}/|I_{\mathrm{feed}}|^2$

Tip: $I_{feed} = I_0\sin(k_0L/2)$ for arbitrary dipole; for $L = \lambda/2$: $I_{feed} = I_0$; for $L = 1.5\lambda$: $I_{feed} = -I_0$

ANTENNA TYPES — FULL EQUATIONS

Arbitrary-Length Dipole (L along $\hat{z}$)

$I(z) = I_0\sin k_0(L/2 - |z|)$ $U(\theta) = \frac{Z_0 I_0^2}{8\pi^2}\left[\frac{\cos(k_0\frac{L}{2}\cos\theta) - \cos(k_0\frac{L}{2})}{\sin\theta}\right]^2$

Half-Wave Dipole ($k_0L/2 = \pi/2$, $\cos(\pi/2) = 0$)

$f(\theta) = \left[\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\right]^2$; $P_{\mathrm{rad}} = \frac{Z_0 I_0^2}{4\pi}(1.219) = 36.56 I_0^2$

1.5λ Dipole ($k_0L/2 = 3\pi/2$, $\cos(3\pi/2) = 0$)

$U(\theta) = \frac{Z_0 I_0^2}{8\pi^2}\left[\frac{\cos(1.5\pi\cos\theta)}{\sin\theta}\right]^2$; $P_{\mathrm{rad}} = 52.74 I_0^2$

$D_{\max} = 2.22$ at $\theta \approx 42.5^\circ$ (NOT at broadside!); $R_{\mathrm{rad}} = 105.5\,\Omega$

Dipole orientation — replace θ with γ:

$\cos\gamma = \hat{d}\cdot\hat{r}$, $\sin^2\gamma = 1 - (\hat{d}\cdot\hat{r})^2$; $\hat{r} = \sin\theta\cos\phi\,\hat{x} + \sin\theta\sin\phi\,\hat{y} + \cos\theta\,\hat{z}$

x-dipole: $f = \frac{\cos^2(\frac{\pi}{2}\sin\theta\cos\phi)}{1-\sin^2\theta\cos^2\phi}$ **y-dipole:** $f = \frac{\cos^2(\frac{\pi}{2}\sin\theta\sin\phi)}{1-\sin^2\theta\sin^2\phi}$

Aperture — Circular (radius $a = D/2$, uniform)

$F(\theta) \propto \frac{2J_1(ka\sin\theta)}{ka\sin\theta}$; 1st null: $ka\sin\theta = 3.832$; 1st SL: -17.6 dB

HPBW= $1.02\lambda/D$; $D_{\max} = 4\pi A_{\mathrm{eff}}/\lambda^2 = 4\pi^2 a^2\eta_{ap}/\lambda^2$

Cosine^n taper $E(r) \propto \cos^n(\pi r/D)$: $n = 0 \to \eta_{ap} = 1.0$; $n = 1 \to 0.72$; $n = 2 \to 0.51$

Aperture — Rectangular ($L_x \times L_y$, uniform)

$F \propto \mathrm{sinc}\left(\frac{\pi L_x\sin\theta\cos\phi}{\lambda}\right)\cdot\mathrm{sinc}\left(\frac{\pi L_y\sin\theta\sin\phi}{\lambda}\right)$

1st SL= -13.5 dB; HPBW= $0.886\lambda/L$; $D = 4\pi L_xL_y/\lambda^2$

DIRECTIVITY, GAIN, BEAM SOLID ANGLE

$D = U_{\max}/U_{\mathrm{avg}} = 4\pi U_{\max}/P_{\mathrm{rad}}$; $\Omega_A = 4\pi/D \approx \Theta_E\Theta_H$

$D \approx 41253/(\Theta_E^\circ\Theta_H^\circ)$ (Kraus — fails for grating lobes or large sidelobes!)

$G = \eta_{\mathrm{eff}}D$; $A_{\mathrm{eff}} = G\lambda^2/4\pi = \eta_{ap}A_{\mathrm{phys}}$

Exact: $D = \frac{4\pi}{\int_0^{2\pi}\int_0^\pi |F(\theta,\phi)|^2\sin\theta\, d\theta\, d\phi/|F_{\max}|^2}$

Aperture efficiency: $\eta_{ap} = \frac{|\iint E\, dA|^2}{A_{\mathrm{phys}}\iint|E|^2\, dA}$ — Schwartz proves max when $E = \mathrm{const}$

RECIPROCITY, EFF. LENGTH, CIRCUIT LINK

Receive pattern $\equiv$ transmit pattern. $A_{\mathrm{eff}} = \lambda^2 D/4\pi$

Open-circuit voltage: $V_{\mathrm{oc}} = \vec{E}\cdot\vec{h}_{\mathrm{eff}} = |E|h_{\mathrm{eff}}\cos\alpha_{\mathrm{pol}}$

Matched load ($Z_L = Z_A^*$): $P_{\mathrm{rec}} = |V_{\mathrm{oc}}|^2/(8R_{\mathrm{rad}}) = S_{\mathrm{inc}}\cdot A_{\mathrm{eff}}$

Example: short dipole ($h_{eff} = L/2$), $|E| = 1$ V/m, $R_{rad} = 20\pi^2(L/\lambda)^2$:

$P = \frac{(L/2)^2}{8\cdot 20\pi^2(L/\lambda)^2} = \frac{\lambda^2}{640\pi^2}$ *W — same as reciprocity result!*

FRIIS, POWER DENSITY, RADAR EQ.

Friis (1-way): $P_r = \frac{P_tG_tG_r\lambda^2}{(4\pi R)^2} = P_tG_t\frac{A_{\mathrm{eff,r}}}{4\pi R^2}$

Power density: $S = P_tG_t/(4\pi R^2)$; $P_r = S\cdot A_{\mathrm{eff}}$

Radar eq. (2-way): $P_r = \frac{P_tG^2\lambda^2\sigma}{(4\pi)^3R^4} = \frac{P_tA_{\mathrm{eff}}^2\sigma}{4\pi\lambda^2R^4}$

Tip: both forms useful — use G form if given gain in dB; use A_{eff} form if given aperture.

Solve for $R_{\max}$: $R_{\max}^4 = \frac{P_tG^2\lambda^2\sigma}{(4\pi)^3\cdot\mathrm{SNR}\cdot k_BTB}$

Thermal noise: $P_N = k_BT_{\mathrm{sys}}B$; SNR= P_r/P_N; $k_B = 1.38\times10^{-23}$ J/K

Example: $P_t = 0.1$ W, $G = 293$, $\lambda = 0.028$ m, $\sigma = 0.2$ m^2, $T = 1000$ K, $B = 1$ kHz: $R_{max} = (P_tG^2\lambda^2\sigma/(4\pi)^3\cdot10\cdot k_BTB)^{1/4} \approx 1.5$ km

POLARIZATION

$\hat{e}_{\mathrm{LCP}} = (\hat{x} + j\hat{y})/\sqrt{2}$; $\hat{e}_{\mathrm{RCP}} = (\hat{x} - j\hat{y})/\sqrt{2}$

Extract component: $E_L = \vec{E}\cdot\hat{e}_L^*$; $|E_L|^2/(2\eta_0)$ = LCP power density

LCP antenna rejects RCP (and vice versa). Linear into circular: 3 dB loss.

Example: $\vec{E} = E_0(3\hat{x} + 2e^{j\pi/4}\hat{y})$; $E_L = \frac{E_0}{\sqrt{2}}(3 - j2e^{j\pi/4}) = \frac{E_0}{\sqrt{2}}(3 + \sqrt{2} - j\sqrt{2})$;

$|E_L|^2 = \frac{E_0^2}{2}(13 + 6\sqrt{2}) \approx 10.74E_0^2$

Polarization loss: $P_{\mathrm{rec}} = S_{\mathrm{inc}}\cdot A_{\mathrm{eff}}\cdot|\hat{e}_{\mathrm{inc}}\cdot\hat{e}_{\mathrm{ant}}^*|^2$

ARRAYS — FULL EQUATIONS & EXAMPLES

Uniform Linear Array (N elements, spacing d, phase α)

$\psi = kd\cos\gamma + \alpha$; $|AF| = \left|\frac{\sin(N\psi/2)}{N\sin(\psi/2)}\right|$ (normalized to 1)

Max when $\psi = 0$; **Nulls** when $N\psi/2 = m\pi$ ($m \neq 0, N, 2N$), i.e. $\psi_{\mathrm{null}} = 2m\pi/N$

Broadside ($\alpha = 0$): beam at $\gamma = 90^\circ$; sidelobe at $3/2\cdot2\pi/N$ from main beam

Ordinary endfire ($\alpha = -kd$): beam at $\gamma = 0^\circ$; $\psi \in [-2kd, 0]$

Hansen-Woodyard ($\alpha = -kd - \pi/N$): $D \approx 1.8\times$ ordinary endfire; first SL≈ -10 dB; $x = \pi$ rule-of-thumb, exact $x \to 2.94$ large N

$N+1$	Conv D	x_{opt}	HW D
5	7.4	1.94	10.6
10	15.3	2.46	24.4
20	31.3	2.71	53.0

Dipole Arrays — Complete Pattern

N-element z-dipole array along x-axis (spacing d, phase $\alpha = 0$ broadside):

$$P(\theta,\phi) = \underbrace{\left[\frac{\cos(\frac{\pi}{2}\cos\theta)}{\sin\theta}\right]^2}_{\mathrm{EF}}\cdot\underbrace{\left[\frac{\sin(N\cdot\frac{\pi d}{\lambda}\sin\theta\cos\phi)}{N\sin(\frac{\pi d}{\lambda}\sin\theta\cos\phi)}\right]^2}_{\mathrm{AF}}$$

E-plane ($\phi = \pi/2$, yz-plane): AF= 1 everywhere $\Rightarrow$ pattern = EF alone (HPBW= 78° for $\lambda/2$ dipole)

H-plane ($\theta = \pi/2$, xy-plane): EF= 1 $\Rightarrow$ pattern = AF alone; HPBW depends on N, d

Tip: d close to $\lambda \Rightarrow$ grating lobes; Kraus estimator fails; use exact 2D integral for true D.

Example: 21-element z-dipole, $d = 0.9\lambda$: $HPBW_H = 2.7^\circ$, $HPBW_E = 78^\circ$, Kraus$\approx$ 196 but exact $D \approx 77$ due to grating lobes!

Half-Wave Dipole Array (x-directed dipoles, x-array)

EF for x-directed half-wave dipole: $EF(\theta,\phi) = \frac{\cos(\frac{\pi}{2}\sin\theta\cos\phi)}{\sqrt{1-\sin^2\theta\cos^2\phi}}$

In E-plane ($\phi = 0$, xz-plane): $EF = \cos(\frac{\pi}{2}\sin\theta)/\cos\theta$

In H-plane ($\phi = \pi/2$, yz-plane): $EF = 1$ (independent of θ) $\Rightarrow$ H-plane pattern = AF only

2D Planar Array ($N_x \times N_y$, spacings d_x, d_y)

$P(\theta,\phi) = |EF|^2\cdot|AF_x|^2\cdot|AF_y|^2\cdot|GPF|^2$

$AF_x = \frac{\sin(N_x\psi_x/2)}{N_x\sin(\psi_x/2)}$; $\psi_x = kd_x\sin\theta\cos\phi + \alpha_x$

$AF_y = \frac{\sin(N_y\psi_y/2)}{N_y\sin(\psi_y/2)}$; $\psi_y = kd_y\sin\theta\sin\phi + \alpha_y$

Example: 5×5 x-dipole array, $d = \lambda/2$, $h = \lambda/4$ above ground: $kd = \pi$; $AF_x = \sin(\frac{5\pi}{2}\sin\theta\cos\phi)/\sin(\frac{\pi}{2}\sin\theta\cos\phi)$; $GPF = 2\sin(\frac{\pi}{2}\cos\theta)$; null at horizon since $\sin(0) = 0$; beam broadside $\theta = 0$.

Scan loss: beam at $\theta_0 \Rightarrow$ HPBW widens by $\approx 1/\cos\theta_0$.

Ground Plane Factor (Image Theory)

Horizontal element at height h: GPF= $2\sin(kh\cos\theta)$ ($\alpha = \pi$ image)

Vertical element at height h: GPF= $2\cos(kh\cos\theta)$ ($\alpha = 0$ image)

Tip: GPF forces null at $\theta = 90^\circ$ for horizontal (sine), max at $\theta = 90^\circ$ for vertical (cos, if $kh\cos 90 = 0$). Check boundary!

Beam Steering

$\alpha_x = -kd_x\sin\theta_0\cos\phi_0$; $\alpha_y = -kd_y\sin\theta_0\sin\phi_0$

Example: steer to $\theta_0 = 25^\circ$, $\phi_0 = 90^\circ$ in y-array with $d = \lambda/2$: $\alpha_y = -\pi\sin 25^\circ \approx -1.327$ rad $\approx -0.423\pi$

APERTURE EFFICIENCY & TAPERING

Schwartz: $|\int AB\, ds|^2 \leq \int|A|^2 ds\cdot\int|B|^2 ds$; equality iff $B = cA \Rightarrow$ uniform max.

Tapering trade-off: lower sidelobes $\leftrightarrow$ wider HPBW, lower η_{ap}, lower D.

Convolution theorem: $E_{\mathrm{total}} = E_1 * E_2 \Rightarrow$ pattern$\propto|F_1|^2\cdot|F_2|^2$

Aperture	SLL	HPBW	η_{ap}
Uniform (1D)	-13.5 dB	$0.886\lambda/L$	1.0
Triangular (1D)	-27 dB	$1.27\lambda/L$	0.75
Uniform circ	-17.6 dB	$1.02\lambda/D$	1.0
Cosine^1 circ	lower	wider	0.72
Cosine^2 circ	much lower	wider	0.51
Quadratic taper	-26 dB	matches unif	≈ 0.50

Triangular = rect$*$rect (each $L/2$). Power pattern = sinc$^4(kL\sin\theta/4)$, where sinc^2 from each rect.

Fraunhofer boundary: $r_0 = 2D^2/\lambda$ (D=max linear dimension, NOT directivity!). At $r \gg r_0$: sinc^2 pattern. At $r \ll r_0$: flat "searchlight" beam with phase ripple.

NOISE: CASCADE, SKY, CABLES

Friis cascade (referred to stage-1 input): $T_{\mathrm{sys}} = T_1 + \frac{T_2}{G_1} + \frac{T_3}{G_1G_2} + \cdots$

First stage dominates!

Lossy cable ($\varepsilon = 10^{-L_{\mathrm{dB}}/10}$, physical temp $T_p = 300$ K):

- Referred to cable input: $T_{\mathrm{cab,in}} = \frac{1-\varepsilon}{\varepsilon}T_p = (L-1)T_p$
- Referred to cable output: $T_{\mathrm{cab,out}} = (1-\varepsilon)T_p$

System (ant $\to$ cable $\to$ receiver): $T_{\mathrm{sys}} = T_{\mathrm{sky}} + \frac{1-\varepsilon}{\varepsilon}T_p + \frac{1}{\varepsilon}T_{\mathrm{rec}}$

LNA at antenna (ant $\to$ LNA $\to$ cable $\to$ stages): $T_{\mathrm{sys}} = T_{\mathrm{sky}} + T_1 + \frac{1-\varepsilon}{\varepsilon}\frac{T_p}{G_1} + \frac{T_2}{G_1\varepsilon} + \cdots$ (dramatic improvement!)

Example: RG-8, 50 m at 400 MHz: $L_{dB} = 5\times1.64 = 8.2$ dB, $\varepsilon = 0.15$. $T_{sys} = 50 + 5.67(300) + 6.67(42) = 2031$ K. Move LNA first: $T_{sys} \approx 120$ K — 17$\times$ better!

RANGE, DOPPLER, ALIASING

True range: $R = c\,t_{\mathrm{true}}/2$; $t_{\mathrm{true}} = t_{\mathrm{obs}} + n\cdot T_{\mathrm{IPP}}$

Max unambiguous range: $R_{\mathrm{unamb}} = c\cdot T_{\mathrm{IPP}}/2$

Doppler: $f_d = 2v_r/\lambda$ (approaching $\to$ positive f_d); $v_r = \lambda f_d/2$

Max unambiguous Doppler/vel: $|f_d| \leq$ PRF/2; $|v| \leq \lambda\cdot$PRF/4

De-alias range (2 IPPs, find smallest $n_1, n_2 \geq 0$): $t_1 + n_1T_1 = t_2 + n_2T_2$; try $n_1 = 0, 1, 2\ldots$ until integer n_2 found.

Example: $IPP_1 = 500\,\mu s$, $t_1 = 300\,\mu s$; $IPP_2 = 400\,\mu s$, $t_2 = 200\,\mu s$: $300 + 500n_1 = 200 + 400n_2 \Rightarrow 4n_2 - 5n_1 = 1 \Rightarrow n_1 = 3, n_2 = 4 \Rightarrow t_{true} = 1800\,\mu s \Rightarrow R = 270$ km

De-alias Doppler (2 PRFs, m can be $\pm$): $f_1 + m_1\mathrm{PRF}_1 = f_2 + m_2\mathrm{PRF}_2$; find smallest $|f_{\mathrm{true}}|$

Example: $PRF_1 = 2000$ Hz, $f_1 = +400$ Hz; $PRF_2 = 2500$ Hz, $f_2 = +900$ Hz: $4m_1 - 5m_2 = 1 \Rightarrow m_1 = -1, m_2 = -1 \Rightarrow f_{true} = -1600$ Hz

OVERSPREAD TARGETS

Overspread if $\tau_{\max}\cdot B_D > 1$: $\tau_{\max} = 2L/c$; $B_D = 4\Omega y f_0/c$; $\tau_{\max}B_D = 8R^2\Omega f_0/c^2$

$f_{\mathrm{thresh}} = c^2/(8R^2\Omega)$ (below = underspread)

Small angle cap: $\theta_{\max} = (c^2/4R^2\Omega f_0)^{1/3}$; $L \approx R\theta^2/2$; $y \approx R\theta$

Example: Mars, $R = 3400$ km, $\Omega = 7.06\times10^{-5}$ rad/s, $f_0 = 2.38$ GHz: $f_{thresh} = 13.8$ MHz; both 2.38 GHz and 430 MHz $\gg f_{thresh} \Rightarrow$ severely overspread. Restrict to $\theta_{max} = 0.226$ rad $\Rightarrow L = 87$ km, $2y = 1530$ km.

PULSE COMPRESSION & CODES

Key metrics: CR= N (baud count); $\Delta R = cT_c/2$; bandwidth $B \approx 1/T_c$; SLL= $20\log_{10}$(peak sidelobe/N) dB

Autocorrelation: $R(k) = \sum_n x[n]x[n+k]$; peak= N; matched filter output=ACF

Barker codes: all sidelobes = ±1; aperiodic ACF ideal. Longest known: $N = 13$, SLL= -22.3 dB.

N	Sequence	SLL
2	$++$	-6 dB
3	$++-$	-9.5 dB
4	$+++-$	-12 dB
5	$+++-+$	-14 dB
7	$+++--+-$	-16.9 dB
11	$+++---+--+-$	-20.8 dB
13	$+++++--++-+-+$	-22.3 dB

Compound code B_{NM}: outer B_N baud$\to$inner B_M; outer$+\to B_M$; outer$-\to\overline{B_M}$; CR= NM. *B_{43}: outer $B_4 = (+,+,-,+)$, inner $B_3 = (+,+,-)$: $(+,+,-,+,+,+,-,-,+,+,+,+,-)$; SLL= -9.5 dB, CR= 12*

Complementary pair (A, B): $R_A(k) + R_B(k) = 0$ for $k \neq 0$; must transmit both and sum.

Build recursively: $A_{n+1} = A_n\|B_n$; $B_{n+1} = A_n\|\overline{B_n}$

$A_1 = (+,+)$, $B_1 = (+,-)$; $A_2 = (+,+,+,-)$, $B_2 = (+,+,-,+)$; $A_3 = (+,+,+,-,+,+,-,+)$

m-sequences (LFSR): p stages $\Rightarrow N = 2^p - 1$; periodic ACF: peak= N at $k = 0 \pmod N$, -1 elsewhere; choose p so $N\tau \leq T_{\mathrm{IPP,max}}$

Mars example: need $T_{IPP} \le 583\,\mu s$ with $\tau = 4\,\mu s$: $p = 7 \Rightarrow N = 127$, IPP$= 508\,\mu s < 583\,\mu s$ ✓

AMBIGUITY FUNCTION

$|\chi(\tau,\nu)| = |\int p(t)p^*(t-\tau)e^{j2\pi\nu t}dt|$

$\tau = 0$ cut: $\approx \mathrm{sinc}(\nu T_{\mathrm{total}})$ (freq response; width$\propto 1/T_{\mathrm{total}}$)

$\nu = 0$ cut: standard ACF (range profile; width$= T_c$ for coded pulse)

Ideal: single thumbtack at $(0,0)$; Woodward theorem: $\iint |\chi|^2 d\tau\, d\nu = 1$ (volume conserved — sharpening one dim widens another)

Barker: great $\nu = 0$ cut; degrades rapidly off-axis (Doppler sensitive)

m-sequence: good periodic ACF; also Doppler vulnerable for large ν

IONOSPHERIC PROPAGATION

Secant Law / MUF: $f_p \approx 9\sqrt{N_{\max}}$ Hz; $\mathrm{MUF} = f_p \sec\psi_i = f_p/\cos\psi_i$

Operate at $f \le$ MUF; $f >$ MUF $\Rightarrow$ wave escapes to space.

Ionospheric geometry (law of sines in Earth-ionosphere triangle):

$$\left(1 + \frac{h'}{R_e} - \cos\theta\right)\csc\theta = \cot\psi_i \;; \quad \theta = \frac{d/2}{R_e} \text{ (rad)}$$

Elevation angle: $\beta = 90^\circ - \theta - \psi_i$ (above local horizon)

Example: $d_{total} = 6760$ km, 2 hops, $h' = 300$ km: $d = 3380$ km, $\theta = 1690/6371 = 0.265$ rad$= 15.2^\circ$; $\cot\psi_i = (1 + 0.0471 - 0.9650)/0.2622 = 0.313 \Rightarrow \psi_i = 72.6^\circ$; $\beta = 90 - 15.2 - 72.6 = 2.2^\circ$. Day ($N_{max} = 10^{12}$): MUF$= 9\times10^6/0.299 = 30.1$ MHz. Night ($N_{max} = 10^{11}$): MUF$= 2.846\times10^6/0.299 = 9.5$ MHz.

Generalized Snell's Law (stratified, $n = n(y)$): $n(y)\sin\theta(y) = n_0\sin\theta_0 =$ const

$\frac{dx}{dy} = \frac{\sin\theta_0}{\sqrt{n^2(y) - \sin^2\theta_0}}$; turning point: $n^2(y) = \sin^2\theta_0$

$\int \frac{du}{\sqrt{a^2-u^2}} = \arcsin(u/a)$ (key integral for ray tracing)

For $n^2 = 1 - y^2$: $y(x) = \cos\theta_0\sin(x/\sin\theta_0)$; max depth $y_{\max} = \cos\theta_0$; emerges at $x = \pi\sin\theta_0$. Reaches $n = 0$ only at $\theta_0 = 0^\circ$.

COHERENT/INCOHERENT INTEGRATION

Coherent (N-pt FFT): SNR gain$= N$; $\Delta f_{\mathrm{bin}} = \mathrm{PRF}/N$; $T_{\mathrm{FFT}} = N\cdot T_{\mathrm{IPP}}$; match Δf_{bin} to target spectral width. Signal power $\propto N^2$, noise power $\propto N$.

Incoherent (K spectra): variance $\propto 1/\sqrt{K}$; $K \le T_{\mathrm{dwell}}/T_{\mathrm{FFT}}$; $T_{\mathrm{dwell}} = \Delta R/v$

Example: $f = 300$ MHz, $\tau = 20\,\mu s$, IPP$= 2$ ms, $v = 40$ m/s, $\Delta f_{spec} = 2$ Hz: $\Delta R = c\tau/2 = 3$ km; $T_{dwell} = 3000/40 = 75$ s; need $500/2^n = 2$ Hz $\Rightarrow n = 8$, $N = 256$; gain$= 10\log 256 = 24$ dB; $-20 + 24 = +4$ dB SNR. $K = 75/0.512 = 146$ spectra for incoherent avg.

RICIAN DISTRIBUTION

$P(|V|) = \frac{|V|}{\sigma^2}\exp\left(-\frac{|V|^2+|V_0|^2}{2\sigma^2}\right)I_0\left(\frac{|V||V_0|}{\sigma^2}\right)$

Low SNR ($|V_0| \ll \sigma$): $I_0 \approx 1 \Rightarrow$ Rayleigh: $P(|V|) = \frac{|V|}{\sigma^2}e^{-|V|^2/2\sigma^2}$

High SNR ($|V_0| \gg \sigma$): $I_0(x) \approx e^x/\sqrt{2\pi x} \Rightarrow$ Gaussian at $|V_0|$ with variance σ^2

Joint to polar: area element $dx\,dy = |V|d|V|d\phi$; integrate out ϕ using $\int_0^{2\pi} e^{c\cos\phi}d\phi = 2\pi I_0(c)$.

MATCHED FILTER

$h(t) = K\cdot F^*(t_0 - t)$ (time-reversed conjugate of transmitted signal, delayed by t_0)

Output $y(t) = f(t) * h(t) =$ cross-correlation of received with transmitted; peak $\propto$ total pulse energy.

Example: $F(t) = +1$ on $[0,\tau]$, -1 on $[\tau, 3\tau]$: $h(t) = -1$ on $[0, 2\tau]$, $+1$ on $[2\tau, 3\tau]$. Peak output $= 1^2\cdot\tau + (-1)^2\cdot 2\tau = 3\tau$ at $t_0 = 3\tau$. Zero at $\Delta = \pm\tau$ (phase cancellation). Piecewise linear ACF with max sidelobe $= -\tau$ at $\Delta = 2\tau$.

ARRAY FACTOR — ADDITIONAL DETAILS

3-element endfire array (max $+x$, null $-x$): $\alpha = -kd$; null at $\phi = \pi$: need $\psi(\pi) = -2kd = 2m\pi/3 \Rightarrow d = m\lambda/6$; primary: $d = \lambda/6$, $\alpha = -\pi/3$.

Pattern $= |\sin(\frac{3}{2}\psi)/\sin(\frac{1}{2}\psi)|$; at $\phi = 0$: $\psi = 0 \Rightarrow AF = 3$ (max); at $\phi = \pi$: $\psi = -2kd = -2\pi/3 \Rightarrow \sin(-\pi) = 0 \Rightarrow$ null ✓

HPBW shortcuts:

- Uniform circular aperture: HPBW$= 1.02\lambda/D$ (rad)
- Uniform rectangular: HPBW$= 0.886\lambda/L$ (rad)
- Triangular (1D): HPBW$= 1.27\lambda/L$ (rad)
- Half-wave dipole: HPBW$= 78^\circ$ (E-plane only)
- Broadside N-element, $d = \lambda/2$: HPBW$\approx 2\lambda/(Nd) = 2/N$ rad

First null location (N-element uniform array): $\sin\theta_{\mathrm{null}} = \lambda/(Nd)$ (broadside)

COMPLETE WORKED EXAMPLES

Aperture Efficiency (Circular, $D = 133\lambda$, $E \propto \cos^n(\pi r/D)$)

$A_{\mathrm{phys}} = \pi(D/2)^2 = 13893\lambda^2$; $D_{\max}(\eta_{ap} = 1) = \pi^2(133)^2 \approx 174577$

$\eta_{ap} = (\mathrm{avg}[n])^2/\mathrm{avg}[2n]$ where $\mathrm{avg}[n] = \frac{8}{\pi^2}\int_0^{\pi/2} x\cos^n x\,dx$: avg[0] $= 1$, avg[1] $= 0.463$, avg[2] $= 0.297$, avg[4] $= 0.172$

$n = 0$: $\eta = 1.0$, $D = 52.4$ dBi; $n = 1$: $\eta = 0.72$, $D = 51.0$ dBi; $n = 2$: $\eta = 0.51$, $D = 49.5$ dBi

HPBW: $\sqrt{4\pi/D}$ rad; $n = 0$: 0.48°; $n = 1$: 0.57°; $n = 2$: 0.68°

ALTAIR Aperture ($a = 25\lambda$, $\lambda = 1$ m, $D = 50\lambda$)

$ka = 50\pi$; $A_{\mathrm{eff}} = \pi(25)^2 = 1963.5$ m^2; $D = 4\pi A_{\mathrm{eff}}/\lambda^2 = 24674$ (43.9 dBi)

HPBW$= 1.02/(50) = 0.0204$ rad$= 1.17^\circ$; 1st null: $\theta = 3.832/(50\pi) = 0.0244$ rad; SL$= -17.6$ dB

$G_{BS} = G_{\max}/2 = 12337$ (40.9 dBi); $A_{BS} = A_{\mathrm{eff}}/2 = 981.75$ m^2 (backscatter, used twice)

SETI Link Budget (Arecibo → Tau Ceti)

$G = 73$ dBi$\rightarrow$ $G_{\mathrm{lin}} = 10^{7.3} \approx 2\times10^7$; $A_{\mathrm{eff}} = G\lambda^2/4\pi = (2\times10^7)(0.12)^2/(4\pi) = 2.29\times10^4$ m^2

$P_r = P_tG_tA_{\mathrm{eff,r}}/(4\pi R^2) = (10^6)(2\times10^7)(2.29\times10^4)/(4\pi(1.04\times10^{17})^2) = 3.36\times10^{-19}$ W

$T_{\mathrm{sys}} = 10 + 38 = 48$ K; $N_0 = k_BT = 6.62\times10^{-22}$ W/Hz; $B = P_r/(\mathrm{SNR}\cdot N_0) = 50.7$ Hz ≈ 50 bit/s

Cost: $t = 40/50 = 0.8$ s per word; $E = 10^6\times0.8 = 8\times10^5$ J$= 0.222$ kWh; cost$=$ \$0.033/word

Key insight: cost is independent of P_t (doubling $P_t \Rightarrow$ doubles $B \Rightarrow$ halves $t \Rightarrow$ same energy)

BACKSCATTER & RADAR CROSS-SECTION

Distributed volume (antenna used twice): $P_r = \frac{P_tA_{\mathrm{eff}}^2\sigma_v\Delta V}{4\pi\lambda^2R^4}$; $G_{BS} = G_{\max}/2$; $A_{BS} = A_{\mathrm{eff}}/2$

Radar detector advantage (1-way vs 2-way): detector $\propto 1/R^2$; radar echo $\propto 1/R^4$.

Police radar example: $P_t = 0.1$ W, $G = 293$, $f = 10.55$ GHz, $\sigma = 0.2$ m^2, $T = 1000$ K, $B = 1$ kHz: $R_{max} = (P_tG^2\lambda^2\sigma/(4\pi)^3\cdot P_{N,min})^{1/4} = 1.5$ km. Detector (1-way, $G_{det} = 4$, $T_{det} = 5000$ K, $B = 10$ kHz): $R_{det} = (P_tA_{eff,rad}G_{det}/(4\pi P_{r,det}))^{1/2} = 294$ km! Earth curvature limits practical range.

QUICK-REFERENCE CONSTANTS & CONVERSIONS

$c = 3\times10^8$ m/s; $k_B = 1.38\times10^{-23}$ J/K; $\eta_0 = 120\pi\,\Omega$; 1 kWh$= 3.6\times10^6$ J

$G[\mathrm{lin}] = 10^{G_{\mathrm{dB}}/10}$; $V[\mathrm{lin}] = 10^{V_{\mathrm{dB}}/20}$; $\varepsilon = 10^{-L_{\mathrm{dB}}/10}$; $\lambda = c/f$

$\int\cos^n\theta\sin\theta\,d\theta = -\cos^{n+1}\theta/(n+1)$; $\int re^{-ar^2}dr = -e^{-ar^2}/(2a)$

Euler: $e^{j\theta} = \cos\theta + j\sin\theta$; $\sin(\pi - x) = \sin x$; $\csc\theta = 1/\sin\theta$; $\cot\psi = \cos\psi/\sin\psi$

$\int_0^\pi \sin^3\theta\,d\theta = 4/3$; $\int_0^\pi \frac{\cos^2(\frac{\pi}{2}\cos\theta)}{\sin\theta}d\theta = 1.219$

dB shortcuts: 3 dB $= 2\times$; 6 dB $= 4\times$; 10 dB $= 10\times$; 20 dB $= 100\times$; 30 dB $= 1000\times$

PRF/IPP: PRF$= 1/T_{\mathrm{IPP}}$; $f_d = 2v_r/\lambda$; $v_{\mathrm{unamb}} = \lambda\cdot\mathrm{PRF}/4$; $R_{\mathrm{unamb}} = cT_{\mathrm{IPP}}/2$

TIPS & COMMON MISTAKES

- D **vs.** D: Fraunhofer $r_0 = 2D^2/\lambda$ — D is aperture size, NOT directivity!
- **Grating lobes**: $d \ge \lambda \Rightarrow$ avoid; Kraus estimator breaks; use exact 2D integral.
- **Phase at main beam**: always check $\psi = 0$ gives your intended direction.
- **Endfire vs broadside**: endfire $\alpha = -kd$ (beam along array axis); broadside $\alpha = 0$ (beam $\perp$ array).
- **Ground plane image phase**: horizontal = opposite phase (π); vertical = same phase.
- **Two-way vs one-way**: radar echo $\propto R^{-4}$; communication $\propto R^{-2}$; detector $\propto R^{-2}$.
- **LNA placement**: always put lowest-noise amplifier first; cable loss before LNA kills SNR.
- **Overspread**: product $\tau_{\max}B_D > 1 \Rightarrow$ cannot map entire target; restrict angular range.
- **Complementary codes**: must transmit and receive BOTH A and B pulses; then sum ACFs.
- **m-sequence length**: $N = 2^p - 1$ (NOT 2^p); IPP$= N\tau_{\mathrm{baud}}$.
- **Compound code SLL**: degrades from inner Barker; SLL not ideal like individual Barker.
- **MUF**: frequency must be $\le$MUF to reflect; $>$MUF $\Rightarrow$ lost to space. MUF increases with $\sec\psi_i$ (oblique paths have higher MUF).
- **Polarization**: always decompose into same basis as antenna; LCP antenna $\perp$ RCP.
- **Aperture efficiency direct**: $\eta_{ap} = |\int E\,dA|^2/(A_{\mathrm{phys}}\int|E|^2dA)$ — can compute directly or via $D\lambda^2/4\pi A_{\mathrm{phys}}$.
- **HPBW** from D: HPBW $\approx \sqrt{4\pi/D}$ rad only for symmetric pencil beams.
- **Scan loss**: steering to θ_0 widens beam: $\mathrm{HPBW}_{\mathrm{steered}} \approx \mathrm{HPBW}_0/\cos\theta_0$.